Aditya Kumar

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EDUCATION

Indian Institute of Technology Delhi (IIT Delhi) *New Delhi, India*

Bachelor of Technology in Electrical Engineering 2015 – 2019

- **ML/AI Coursework:** Machine Learning, Advanced ML, Deep Learning, Computer Vision, NLP, Image Processing
- **Core CS/Math:** Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Digital Signal Processing

EXPERIENCE

Posha *Bengaluru, India*

Lead AI Engineer Oct 2023 – Present

- **Search & Recommendation System:** Architected hybrid search engine combining BM25 with GIST semantic embeddings supporting 700+ daily queries across 1,000+ recipes. Deployed FAISS-based retrieval with sub-100ms latency, achieving **30% gain in NDCG@5**, **20% lift in conversions**, and **70%** success on fuzzy queries. Built personalized recommender using collaborative filtering with cold-start solution via preference-based onboarding.
- **Generative AI & Recipe Customization:** Designed and implemented an automated recipe-adaptation pipeline leveraging **DeepSeek-R1** for controlled synthetic data generation with human-in-the-loop validation. Fine-tuned **Qwen-3** using **SFT** and **GRPO** to produce structured JSON outputs with self-reflective reasoning, enabling autonomous recipe modifications and customizations such as ingredient deletion, substitution, and quantity adjustments, grounded in device constraints and ingredient semantics.
- **Universal Frying Model (UFM):** Fine-tuned **Swin Transformer** for real-time doneness prediction replacing fixed timers. Achieved **0.025 MAE** with 100ms latency, enabling adaptive cooking across diverse ingredients.
- **Food Segmentation & Dispense Localization:** Implemented a **SegFormer**-based semantic segmentation system achieving **97% mIoU** across pan, food, and robotic-arm classes. Developed a dispense-localization module (**90% IoU**) to accurately identify the region of the image where a new ingredient is dispensed; restricting inference to this localized region improved ingredient classification to **82% F1** using Swin Transformers. Deployed the end-to-end pipeline on CPU via ONNX with a latency of 220 ms.
- **Vision-Based Quality Auditors:** Adapted **YOLOv8** for blur, splatter, and occlusion detection, achieving **99% F1**. Integrated **SegFormer**-based splatter segmentation (**69.8% IoU**) and lump detection (**88% IoU** for noodles, meat, atta, etc.), enabling automated pause-and-recovery actions to prevent contamination and cooking failures. Optimized deployment via **ONNX**, achieving **50–120 ms** end-to-end latency.

Flipkart *Bengaluru, India*

Applied Scientist II Jan 2020 – Sep 2023

- **Competitive Intelligence Platform (CIP):** Built end-to-end NLP analytics system ingesting feedback from 5 social platforms across 11 Indic languages. Developed multilingual translation pipeline using Samanantar dataset and synthetic code-mixed data generation. Fine-tuned **T5** model for multi-task learning: Targeted-ABSA (**84.3% F1**), span extraction (**84.7% F1**), and product identification (**88.5% F1**).
- **Business Impact:** Engineered regression ensemble forecasting **NPS** with **1% MAPE** and **Market Share** across 8 BUs with **1.5% MAE**. Integrated VoC metrics into CXO dashboards enabling data-driven brand strategy and competitive benchmarking. Adopted organization-wide for weekly tracking.
- **FashionAI (Generative Design):** Fine-tuned **Stable Diffusion** using Masked Textual Inversion and LoRA for attribute-controlled fashion generation. Achieved **75% attribute adherence** (50% improvement over baseline), enabling 4-way attribute combinations. Unlocked automated catalog generation and new revenue streams.

Samsung Research Institute *Noida, India*

Engineer I Jun 2019 – Dec 2019

- **Gesture Recognition:** Developed a sensor-free hand-gesture recognition system for AR/VR using only RGB input. Trained a **MobileNet–DenseASPP** model achieving **98% IoU** for hand segmentation, followed by a **DenseNet**-based gesture classifier reaching **90% accuracy**. Demonstrated real-time performance with reliable tracking from **20 feet**, securing **3rd place** in the Samsung ML Competition.

- **Photorealistic Compositing:** Trained Conditional GAN (**Pix2Pix**) for realistic image composition and CNN in unsupervised setting to model human realism perception, filtering synthetic COCO composite dataset for quality.

Biomedical Imaging Lab, IIT Delhi — *New Delhi, India*

Undergraduate Researcher Aug 2018 – May 2019

- Collaborated with Medanta Hospital for automated liver and lesion segmentation from CT scans. Trained H-DenseUNet achieving **96% Dice** (liver) and **85% Dice** (lesion) on LiTS dataset.
- Developed SVM classifier with radiologist-supervised handcrafted features, boosting patient-level classification (Healthy vs. Cancerous) by **20%** and slice-level accuracy by **8%**.

PUBLICATIONS

- **Privacy-Preserving Compressive Sensing for Image Recognition**
  Developed data-adaptive framework enabling inference from compressed measurements without image reconstruction. Replaced block-based sampling with learned encoder focusing on semantic regions, reducing data by 50% while preserving accuracy. Suitable for privacy-sensitive applications like elderly fall detection.
- **Estimating Customer Experience from Social Media for Competitive Intelligence**
  Presented system for implicit customer experience estimation from Twitter, Instagram, LinkedIn, and Facebook. Used advanced NLP to tag feedback by marketplace, aspect, and sentiment. Enabled CXOs to track market share, detect trends, and predict NPS for data-informed strategy.
- **Multi-Stage Deep Learning for Voice of Customer Insights**
  Introduced pipeline extracting actionable VoC insights from social media in English, Hindi, and Hinglish. Leveraged real-time feedback to identify customer pain points and product experiences, providing scalable solution for enhancing customer satisfaction.

PATENTS

- **Countertop Cooking Robot** — *PCT/US2024/061909 | Issued Jul 2025*
  Countertop appliance utilizing ML/CV models for automatic meal preparation, featuring micro-dispensing pods, automated stirring, and thermal control logic.
- **Managing Sensor Operation with Respect to Activity** — *202011042598 | Issued Jan 2024*
  Method for optimizing sensor usage in computing devices by detecting activity and inferring values from sensor subsets based on learned relationships.
- **Determining Market Share of an Organization** — *202141036284 | Filed Aug 2021*
  System for predicting market share using VoC data, internal/external metrics, and pre-trained datasets with feature constraints.
- **Generation of User Feedback Data** — *202141010012 | Filed Mar 2021*
  Real-time NLP pipeline for extracting and categorizing social media posts to generate actionable user feedback based on sentiment analysis.

TECHNICAL SKILLS

- **Languages:** Python, Java, LaTeX
- **Deep Learning:** PyTorch, PyTorch Lightning, Keras, vLLM, LangChain
- **Generative AI:** Stable Diffusion, Transformers, LLMs, Fine-tuning (SFT, LoRA, GRPO), Prompt Engineering
- **Computer Vision:** OpenCV, YOLO, SegFormer, Swin Transformer, ViT, UNet, DenseASPP, GANs
- **NLP:** T5, BERT, Targeted-ABSA, Machine Translation, Hybrid Search, Spell Correction
- **ML/Data Science:** Pandas, NumPy, Scikit-Learn, XGBoost, FAISS, Recommender Systems, BM25
- **Tools & Deployment:** Git, Docker, Linux, FastAPI, ONNX, AWS, Edge Deployment, Model Optimization
- **Leadership:** Cross-functional Collaboration, Technical Mentorship, Ownership, Bias for Action